

A 1:2 step up transformer.

The transformer is simply two ordinary inductors coupled together using the 'K' coupling statement.

The syntax for this statement is as shown in the schematic where the K defines the statement, the list of the inductors, La, Lb, Lc, etc., define the inductors to be coupled and the number at the end is the coupling coefficient or coupling factor. This defines how closely coupled the inductors are to each other.

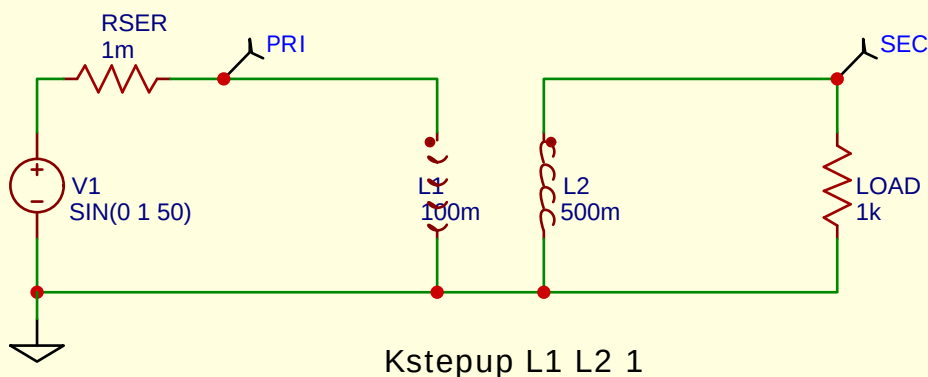
A value of 1 defines perfect coupling whilst 0 defines zero coupling between them.

In practice values of between about 0.9 and 0.99 are achievable in real transformers. Low coupling values may be used for wideband tuned filters built using coupled inductors.

The dotted inductors are used to clearly show the phasing of the windings. The dotted ends are in phase.

Note that a negative coupling coefficient value can be used but it inverts the phase relationship between the primary and secondary windings. It is recommended not to do this as it makes interpreting the phase dots difficult and can cause serious problems in multiwinding coupled inductor circuits.

Notice that a small resistance must be included between the voltage source and the primary inductance to stop spice throwing an error about a loop of voltage sources or a voltage source and an inductor. Basically it is not allowed to place any combination of voltage sources or inductors in parallel with each other without some finite resistance between them.

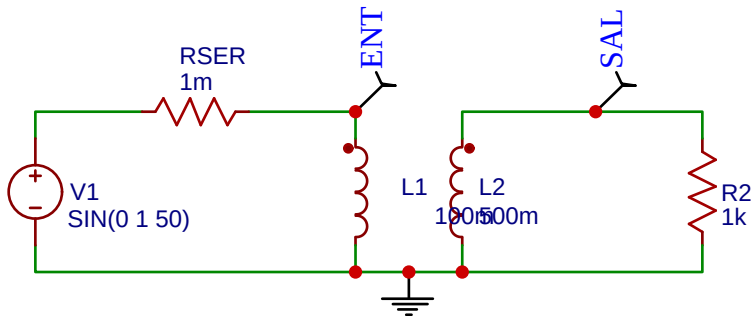


This is the 'K' coupling statement.

It is inserted into the schematic as text and then do:

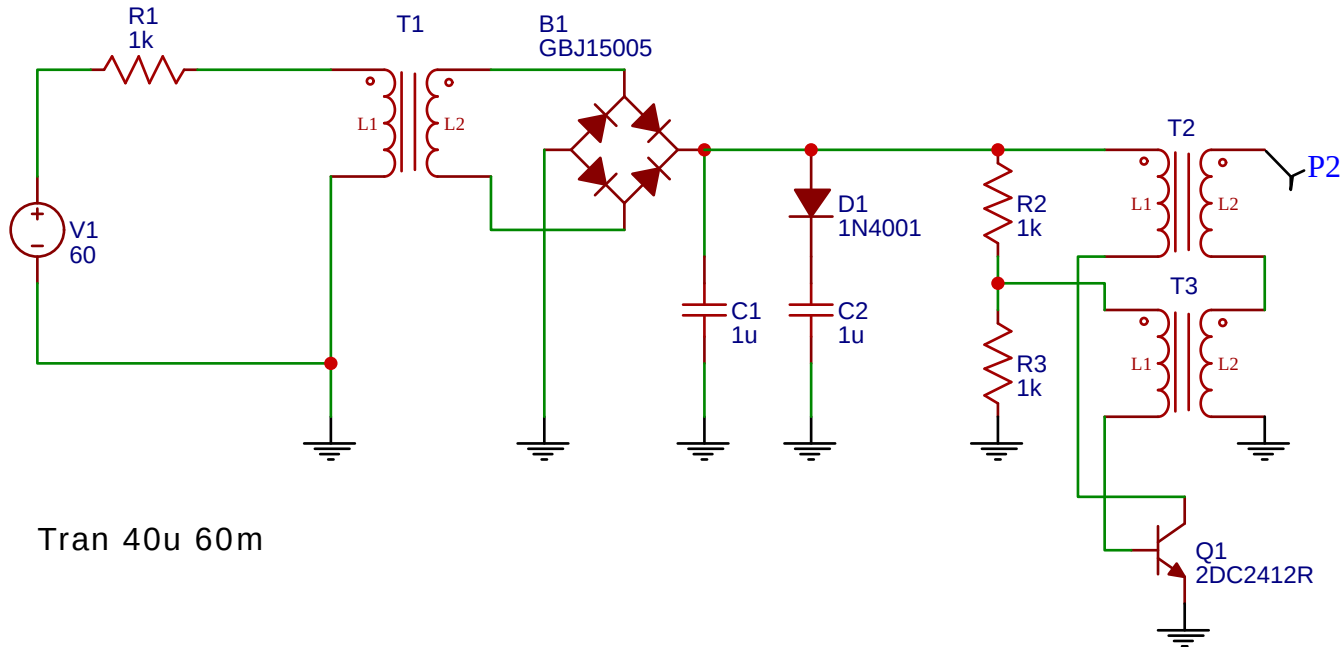
Properties > Text Attributes > Text type > spice

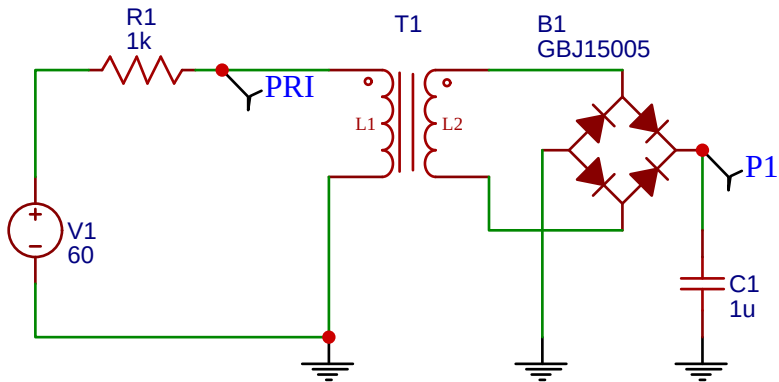
Tran 40u 40m



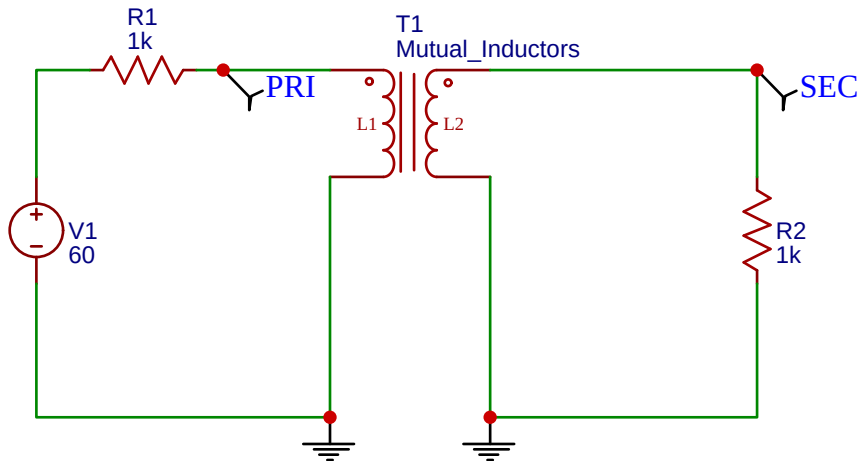
Kstepup L1 L2 1

Tran 40u 40m

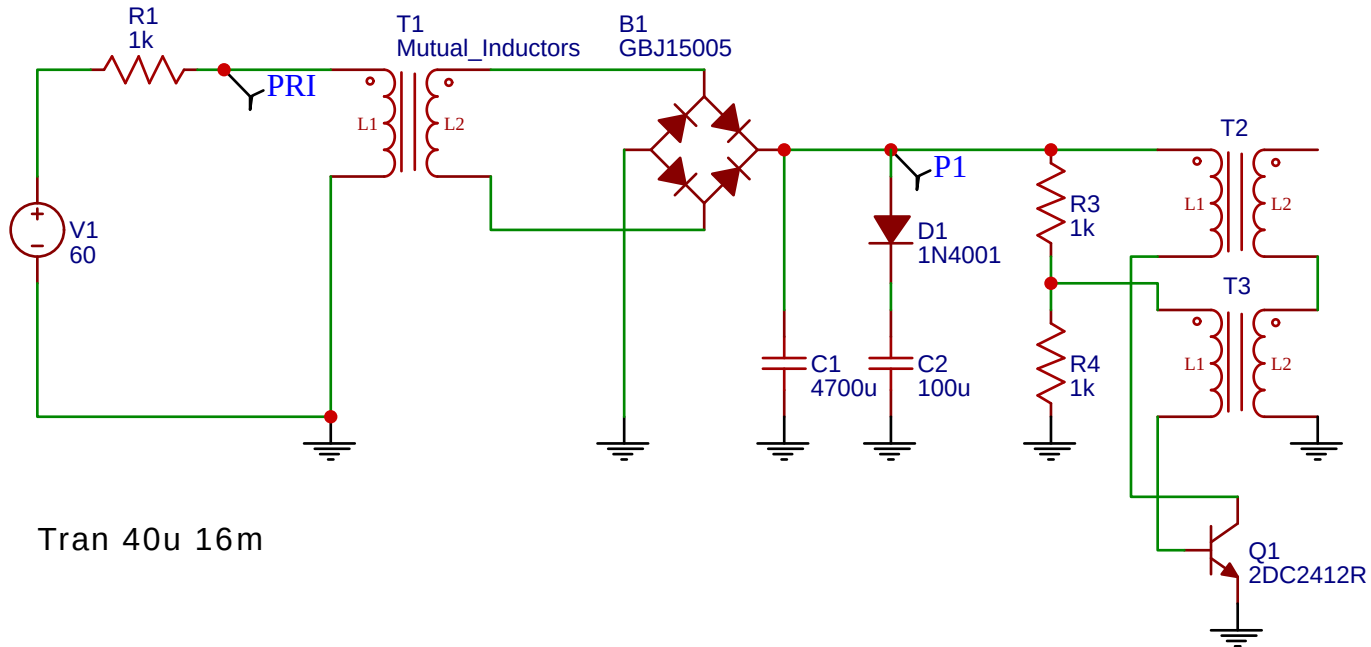


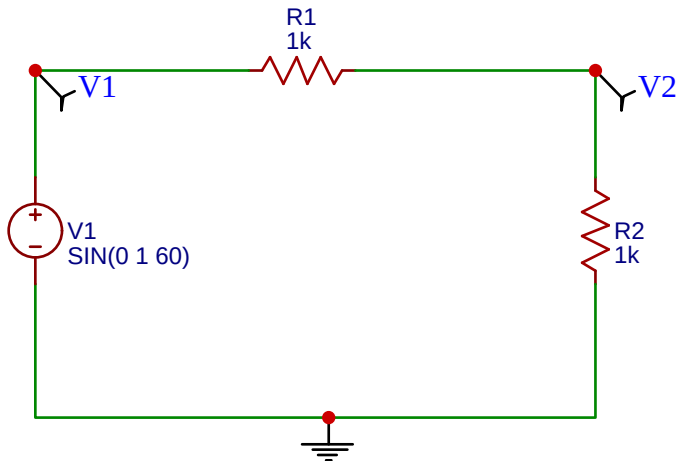


Tran 40u 30m

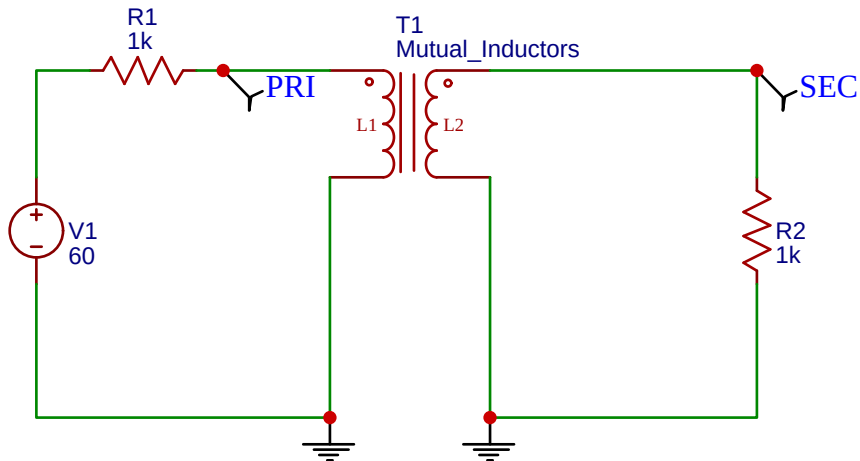


Tran 40u 12.7m

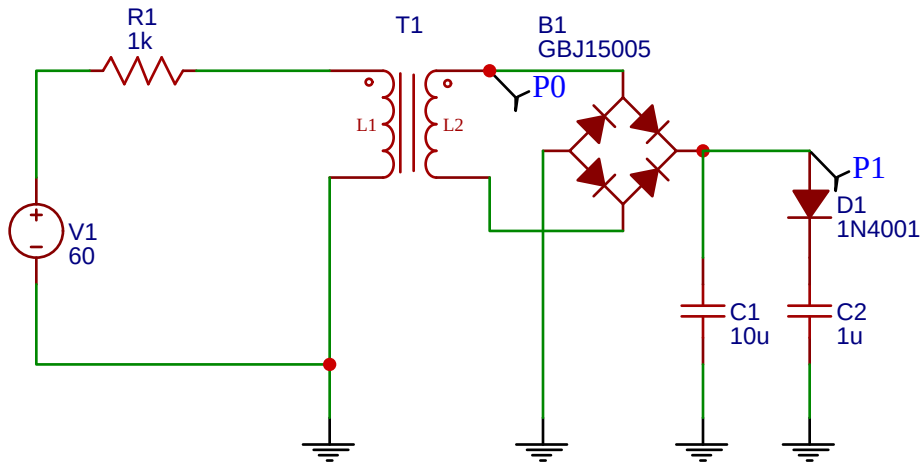




Tran 40u 60m



Tran 40u 12.7m



Tran 40u 60m